

$$n = 4, \boxed{K=17} = \underline{\underline{16^{th}}}$$

$$\boxed{24}$$

$$\begin{array}{cccc} 1 & 2 & 3 & 4 \\ \hline 0 & 1 & 2 & 3 \end{array}$$

$$16/6 = 2$$

$$16 \div 6 = \textcircled{4}$$

$$\underline{1} + \binom{3!}{2, 3, 4} \Big] \underline{6} \quad (0-5)$$

$$\underline{2} + (1, 3, 4) \Big] 6 \quad (6-11)$$

$$\textcircled{3} + (1, 2, 4) \Big] \textcircled{6} \quad (\underline{12-17})$$

$$\underline{4} + (1, 2, 3) \Big] 6 \quad (18-23)$$

$$\underline{\underline{24}}$$

$$1 \ 2 \ 3 \ 4 \rightarrow (0)$$

$$4 \ 3 \ 2 \ 1 \quad \text{---} \quad 23^{th}$$

$$\begin{array}{c} 3 \\ \hline \uparrow \end{array} \quad \text{---} \quad \text{---} \quad \text{---}$$

$$n = 4, K = 17 = 16^{th}$$

$$\boxed{24}$$

$$(1, 2, 3, 4)$$

$$\begin{matrix} & & & 2 \\ 0 & 1 & 2 & 3 \end{matrix}$$

$$\{1, 2, 4\}, K = \frac{4}{2} = 2$$

$$3! = 6$$

$$4 \div 2 = 0$$

$$\{1, 2\} \quad K = 0$$

$$4, 3, 2, 1 \quad \dots \quad 23^{th}$$

$$\begin{matrix} 3 & 4 & & \\ \hline & \uparrow & & \end{matrix}$$

$$① \quad 1 \quad \{2, 4\} \quad] \quad 2 \quad (0-1)$$

$$① \quad 2 \quad \{1, 4\} \quad] \quad 2 \quad (2-3)$$

$$\Rightarrow ② \quad 4 \quad \{1, 2\} \quad] \quad 2 \quad (4-5)$$

$$\underline{\quad\quad\quad}$$

$$6$$

L18. K-th Permutation Sequence | Leetcode

2.40

$$n = 4, K = 17 = 16^{th}$$

$$\boxed{24}$$

$$(1, 2, 3, 4)$$

$$\begin{matrix} & & & \downarrow \\ 0 & 1 & 2 & 3 \end{matrix}$$

$$\{1, 2, 4\}, K = \frac{4}{2} = 2$$

$$3! = 6$$

$$4 \div 2 = 0$$

$$\{1, 2\}$$

$$2! = 2$$

$$\{2\} \quad K = 0$$

$$K = 0 \div 1 = 0$$

$$K = 0 \div 0 = 0$$

$$(3, 4, 1, 2)$$

$$\Rightarrow (1) + \{2\} \quad \downarrow \quad (0-0)$$

$$2 + \{1\} \quad \downarrow \quad (1-1)$$

$$2$$

$$2 + \{ _ \}$$

L18. K-th Permutation Sequence | Leetcode

2.40

$$n = 4, K = 17 = 16^{th} \quad k = 16 / 6 = 2$$

$$(4!) = 24$$

$$3! = 6$$

$$2! = 2$$

$$1! = 1$$

$$(1, 2, 3, 4)$$

✓₃

$$\{1, 2, 4\}, K = (4) / 2 = 2 \quad \checkmark_4$$

$$3! = 6$$

$$4 \div 2 = 0$$

$$\{1, 2\}$$

$$K = 0 / 1 = 0 \quad \checkmark_1$$

$$K = 0 \div 0 = 0$$

$$2! = 2$$

$$\{2\}$$

$$K = 0 \quad \checkmark_2$$

$$(3, 4, 1, 2)$$

16th

$$] 6$$

$$] 6$$

$$] 6 \leftarrow 16 \div 6 = 2$$

$$] 6$$

$$] 2 \quad 0-1$$

$$] 2 \quad 2-3$$

$$] 2 \quad \checkmark (4-5)$$

$$] 1$$

$$] 1$$

$n = 4, K = 17 \Rightarrow 16^{th}$
 $k = \frac{16}{6} = 2$
 $16 \div 6 = 2$
 $TC \rightarrow O(N)$
 $= O(N)$
 $SC = O(N)$

$(4!) = 24$
 $(3!) = 6$
 $(2!) = 2$
 $(1!) = 1$

$\{1, 2, 3, 4\}$
 $K = \frac{4}{2} = 2$
 $4 \div 2 = 0$
 $3! = 6$
 $2! = 2$
 $1! = 1$

$\{2\}$
 $K = 0$
 $ans = 340$

$ans = 340$

C++ Autocomplete

```

1 class Solution {
2 public:
3   string getPermutation(int n, int k) {
4     int fact = 1;
5     vector<int> numbers;
6     for(int i = 1; i < n; i++) {
7       fact = fact * i;
8       numbers.push_back(i);
9     }
10    numbers.push_back(n);
11    string ans = "";
12    k = k - 1;
13    while(true) {
14      ans = ans + to_string(numbers[k / fact]);
15      numbers.erase(numbers.begin() + k / fact);
16      if(numbers.size() == 0) {
17        break;
18      }
19      k = k % fact;
20      fact = fact / numbers.size();
21    }
22    return ans;
23  }
24 };
  
```

2. What is k ?

k tells us:

Which permutation are we looking for?

Initially:

$k = 17$

But the code does:

```
</> C++  
  
k = k - 1;
```

so internally:

$k = 16$

because we're using 0-based indexing.

Think:

```
1st permutation → index 0  
2nd permutation → index 1  
...  
17th permutation → index 16
```

Significance

k tells us:

Where is our desired permutation inside the current block of permutations?

+ |



Think



3. What is $k / \text{fact}!$?

3. What is `k / fact`?

This is the most important one.

```
</> C++  
k / fact
```

tells us:

Which number should I choose next?

Example:

```
k = 16  
fact = 6
```

Therefore:

```
16 / 6 = 2
```

So take:

```
numbers[2] = 3
```

Why?

Because the blocks are:

```
6 permutations each  
1 -> [0 - 5]  
2 -> [6 - 11]  
3 -> [12 - 17]  
4 -> [18 - 23]
```

Our index is `16`.

It falls in the `3` block.

Therefore:

```
k / fact  
= 16 / 6  
= 2
```

which gives index `2`.

Significance

+ |

Think

Permutation Sequence :-

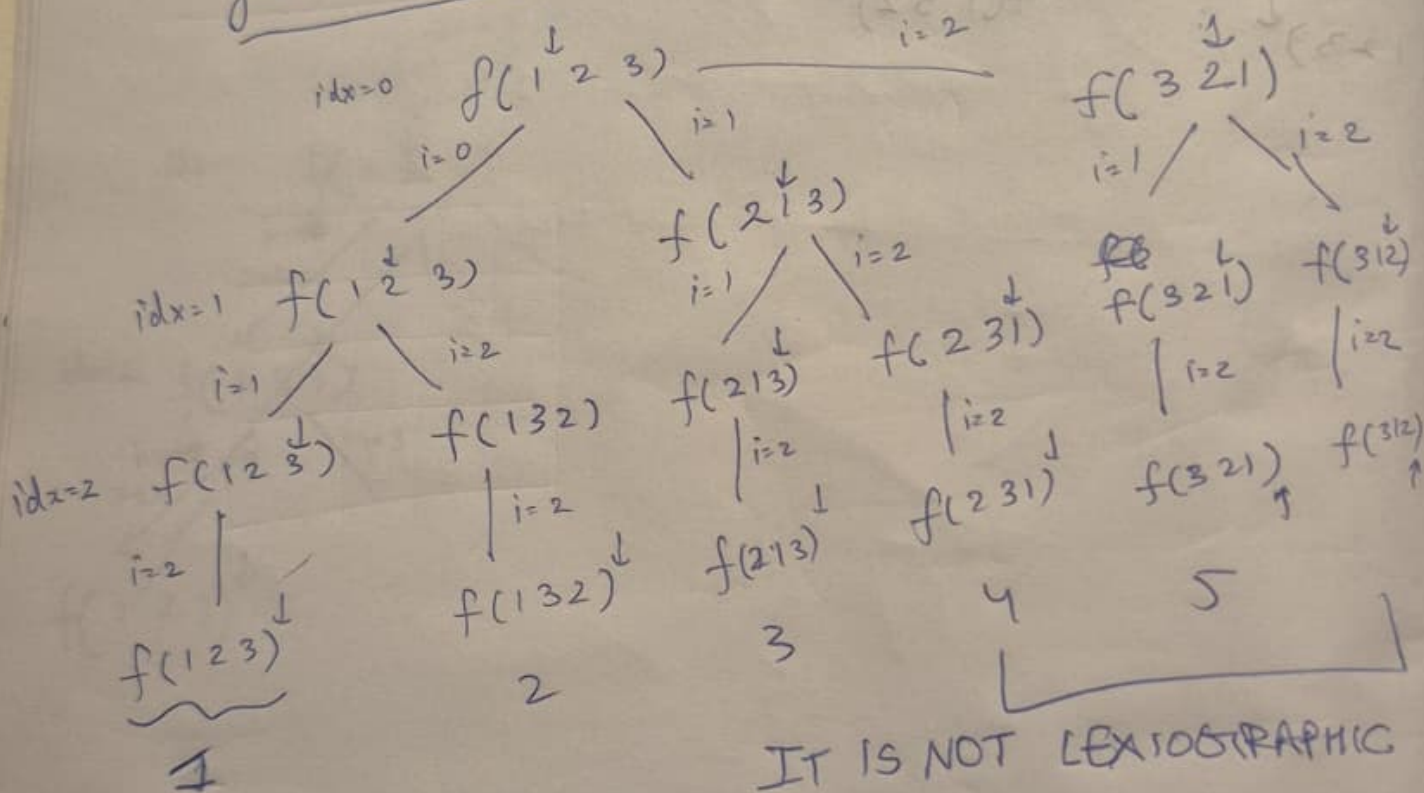
 $n=3$

Use Vis / freq
method → BRITE

1 2 3
1 3 2
2 1 3
2 3 1
3 1 2
~~3 2 1~~

$1\ 2\ 3$
 $1\ 3\ 2$
 $k=3$ $2\ 1\ 3$
 $2\ 3\ 1$
 $3\ 1\ 2$
 $3\ 2\ 1$

find All permutations & keep to counter



Subset Sums	For loop	Palindrome Partitioning	Permutations → for freq/vis
	↑ ChatGPT Image	Permutation Sequence	Subset I, II
		Brute Optimal	All Subsequence → for pick/not pick
			3 types → print 1, count, Good Combination Sum I, II

~~n=4~~ $n=4, k=17$

fact = $1 * 2 * 3 = 6$

numbers [1, 2, 3, 4]

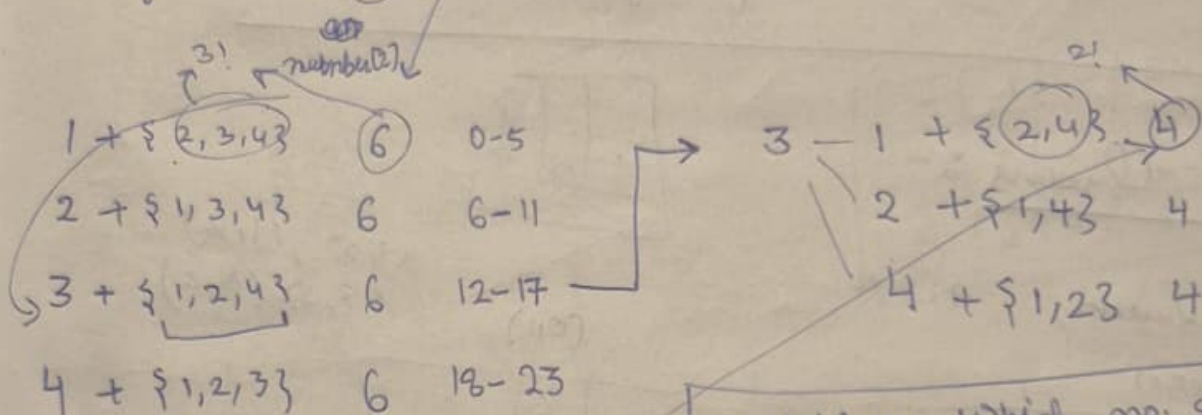
ans = 3

$k=16$

numbers [1, 2, 4]

ans = 3

$k / \text{fact} = 16 / 6 = 2$



$k = k \% \text{fact} = 16 / 6 = 4$

fact = fact / numbers.size()
 = $6 / 3 = 2$

• k / fact : Which no. should I choose next?

$16 / 6 = 2$ numbers (2) = 3

• ~~next~~

• $k \% \text{fact}$:

• fact: How many permutations are inside one block for current position?

1 + {2, 3, 4}
 $3! = 6$

• k: Which permutation are we looking for?

$k=16 \rightarrow$

$k=16$, numbers.size = ~~16~~ 4, fact = 6. ϕ $1 + \underline{\underline{[---]}}$

$k=4$, $[---] = 3$, fact = $6/4 = 2$ $[---]$

$k=2$, $= 2$, fact = $2/2 = 1$ $[-]$

$k=1$, $= 1$, fact = $1/1 = 1$ $[-]$

fact = fact / numbers.size()